MTH357

Examination - January Semester 2025

Applied Regression Analysis I

Thursday, 17 April 2025 **04:00 pm - 06:00 pm**

Time allowed: 2 hours

INSTRUCTIONS TO STUDENTS:

1. This examination contains **FOUR (4)** questions and comprises **EIGHT (8)** printed pages (including cover page).
2. You must answer **ALL** questions.
3. All answers must be written in the answer book. Marks will only be awarded if FULL working is shown.
4. This is an **Open-book** examination.

At the end of the examination.
Please ensure that you have written your examination number on each answer book used.

Failure to do so will mean that your work cannot be identified.
If you have used more than one answer book, please tie them together with the string provided.

THE UNIVERSITY RESERVES THE RIGHT NOT TO MARK YOUR SCRIPT IF YOU FAIL TO FOLLOW THESE INSTRUCTIONS.

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Answer all questions. (Total: 100 marks)

Question 1

The following dataset comprises of the dependent variable Y and the independent, indicator variable X.

	X	Y
1	1	4.4
2	1	13.5
3	1	-1.3
4	1	9.8
5	1	11.3
6	1	9.6
7	1	11.1
8	1	6.9
9	1	19.7
10	1	12.4
11	0	4
12	0	2.8
13	0	0.3
14	0	-2.1
15	0	-1.3
16	0	2.2
17	0	1.9
18	0	-1
19	0	-1.6
20	0	3.8
21	0	-4.1
22	0	3.6
23	0	9.9
24	0	2.4
25	0	7.2

Table Q1

A simple linear regression model was applied to analyze if there is a relation between X and Y.

(a) Estimate the coefficients β_0 and β_1 and interpret the point estimates.

(5 marks)

(b) Given that $s = 4.534$, compute $s(\hat{\beta}_0)$ and $s(\hat{\beta}_1)$, and conduct statistical hypothesis testing to assess if the coefficients are significant at $\alpha = 0.05$.

(8 marks)

(c) Determine the prediction intervals of the regression equation.

(6 marks)

(d) One of your classmates felt that this dataset is too "simple" to require using Simple Linear Regression for analysis. She suggested that you may just use a two sample t-test to determine if X and Y are related. Let Y_1 be the group where $X = 1$ and Y_2 be the group where $X = 0$.

(i) Show how we may use two sample t-test with equal variance to test the hypothesis

$$H_0 : \mu_1 = \mu_2 \text{ against } H_1 : \mu_1 \neq \mu_2$$

Use this information to help you:
Sample standard deviation of $Y_1 = 5.609357$
Sample standard deviation of $Y_2 = 3.679997$

Recall that the t-test statistic is

$$t = \frac{\bar{X}_1 - \bar{X}_2}{S_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}},$$

where S_p^2 is the pooled Variance of the two samples

$$S_p^2 = \frac{S_1^2 (n_1 - 1) + S_2^2 (n_2 - 1)}{(n_1 + n_2 - 2)}$$

(6 marks)

(ii) Calculate the p-value of the test statistic of the two sample t-test and compare with the p-value of the $\hat{\beta}_1$ coefficient. Explain your observations.

(5 marks)

Question 2

The Global F-test (ANOVA) for $H_0 : \beta_1 = 0 \; vs \; H_1 : \beta_1 \neq 0$ is mathematically equivalent to the two-tailed t-test for simple linear regression.

(a) In the Global F test, the sum of squares due to regression is usually presented in this form:

$$SSR = \sum \left(\widehat{y_i} - \bar{y}\right)^2$$

Show that SSR can also be expressed as $SSR = b_1^2 \sum \left(x_i - \bar{x}\right)^2$

Hint: The following equations are useful for this proof:

$$\widehat{y_i} = b_0 + b_1 x_i,$$
$$b_0 = \bar{y} - b_1 \bar{x},$$

where b_0 and b_1 are the unbiased estimators for β_0 and β_1 respectively.

(6 marks)

(b) Show that the test statistic of the two-tailed test for $H_0 : \beta_1 = 0 \; vs \; H_1 : \beta_1 \neq 0, t = \dfrac{b_1}{s\left(b_1\right)}$ is equivalent to the test statistic of the F test of the same hypothesis, $F = \dfrac{MSR}{MSE}$, for the Simple Linear Regression.

Hint: The MSE of the Single Linear Regression is $\dfrac{SSE}{n-2}$, and the Variance of b_1 is $\dfrac{MSE}{\sum (x_i - \bar{x})^2}$.

(6 marks)

(c) Critically comment, with counter-examples if necessary, on the following statements on the coefficient determination, $R^2 = SSR/SST$.

(i) A high R^2 indicates that useful predictions can be made.

(ii) A high R^2 indicates that the estimated regression line is a good fit.

(iii) A low R^2 indicates that X and Y are not related.

(iv) The R^2 can be interpreted as the proportionate reduction of total variation associated with the use of the predictable variable X.

(8 marks)

Question 3

The next two questions will be based on the same dataset.

A group of researchers are interested in understanding how certain variables might affect a person's High Density Lipoprotein (HDL) cholesterol level. HDL cholesterol is known as the "Good cholesterol" and high HDL levels are associated with lower risk for cardiovascular disease.

A sample of 120 people were taken, and information on their Age (in years), Sex (Male = 0, Female = 1), and Weight (Mass) (in Kg) were collected. Each person will then do a blood test to determine their HDL level, measured in mmol/L.

Values of 1.6 mmol/L and above is generally considered good for both sexes, while values lower than 1.3 mmol/L and 1.0 mmol/L are considered risk factors for Women and Men respectively.

A multiple linear regression model with HDL as the dependent variable (y), and Age (x_1), Sex (x_2) and Weight (x_3) as predictor variables was created to analyse the data.

The regression model used is: $y_i = \beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i} + \beta_3 x_{3i} + \varepsilon_i$.

(a) The following is an incomplete table of the results of the regression. Fill in all the empty cells and briefly comment on the results.

	Coef	**Std err**	**t-value**	**P-value**
Intercept		0.317498	6.498	
Age	-0.012041	0.004697		
Sex	0.205875			0.30
Weight	-0.006790		2.05	

Table Q3(a)

Residual standard error: 0.3411 on 116 DF
Multiple R-Squared: 0.2268
F-statistic: 11.34 on 3 and 116 DF, p-val: 1.406e-06

(8 marks)

(b) Interpret the coefficients in the regression model.

(4 marks)

(c) The following is a plot of the Externally Studentized Residuals against $\hat{y}$. Conduct a residual analysis to verify if the assumptions of the regression model were met. Comment on the plot and suggest possible next steps in the regression analysis.

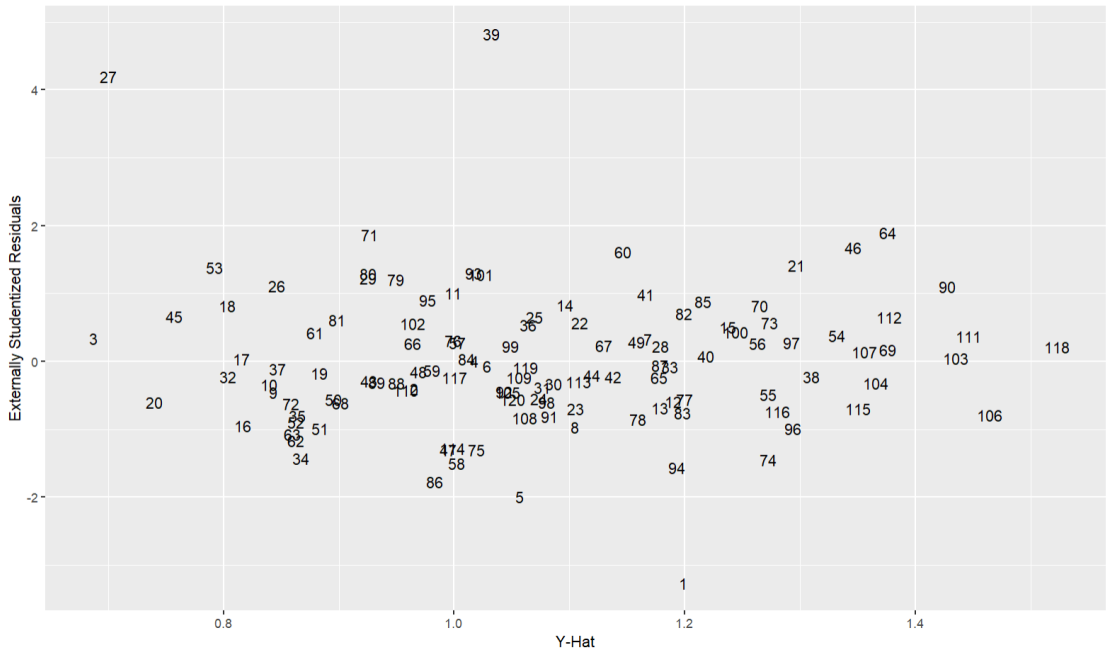


Figure Q3(c)

(4 marks)

(d) To improve on the model fit, we will try two other methods. Using the R outputs, assess how the additional new variable affect model fit and interpretation of the model. Make a recommendation on whether the new variable should be added.

(i) Interaction term between Sex and Weight.

```
Residuals:
     Min       1Q   Median       3Q      Max
-1.04827 -0.21797 -0.03131  0.16353  1.54530

Coefficients:
                 Estimate Std. Error t value Pr(>|t|)
(Intercept)      1.802408   0.428346   4.208 5.13e-05 ***
age             -0.012121   0.004702  -2.578   0.0112 *
sex              0.579843   0.419086   1.384   0.1692
weight          -0.003087   0.005033  -0.613   0.5408
I(sex * weight) -0.005623   0.006193  -0.908   0.3659
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 0.3414 on 115 degrees of freedom
Multiple R-squared:  0.2323,    Adjusted R-squared:  0.2056
F-statistic: 8.699 on 4 and 115 DF,  p-value: 3.596e-06
```

Figure Q3(d)(i)

(5 marks)

(ii) Quadratic term for Weight.

```
Residuals:
     Min       1Q   Median       3Q      Max 
-1.20224 -0.20945 -0.03898  0.16120  1.50566 

Coefficients:
              Estimate Std. Error t value Pr(>|t|)    
(Intercept)  3.0713341  0.5842042   5.257 6.83e-07 ***
age         -0.0126396  0.0046433  -2.722  0.00750 ** 
sex          0.2213000  0.0764176   2.896  0.00453 ** 
weight      -0.0412953  0.0171267  -2.411  0.01748 *  
I(weight^2)  0.0002872  0.0001405   2.044  0.04320 *  
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 0.3365 on 115 degrees of freedom
Multiple R-squared:  0.2539,	Adjusted R-squared:  0.228 
F-statistic: 9.784 on 4 and 115 DF,  p-value: 7.564e-07
```

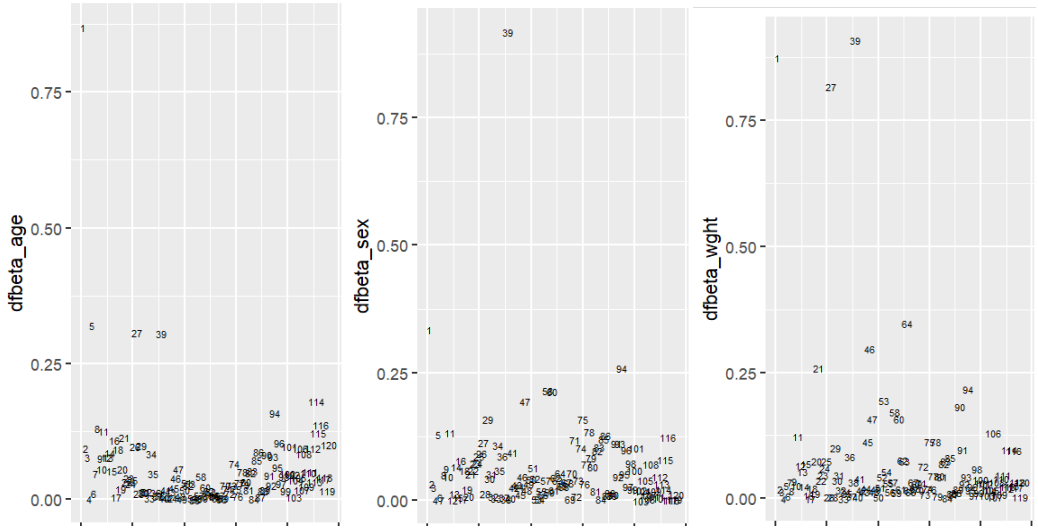
Figure Q3(d)(ii)

(4 marks)

Question 4

This question is a continuation from the previous, using the same dataset. The regression model used is: $y_i = \beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i} + \beta_3 x_{3i} + \varepsilon_i$.

The output from influence.measures() in R is plotted and labelled with the sample number for your convenience. The top three are the DFBETAS for Age, Sex and Weight respectively. The bottom three are the DFFITS and Cook’s Distance respectively.



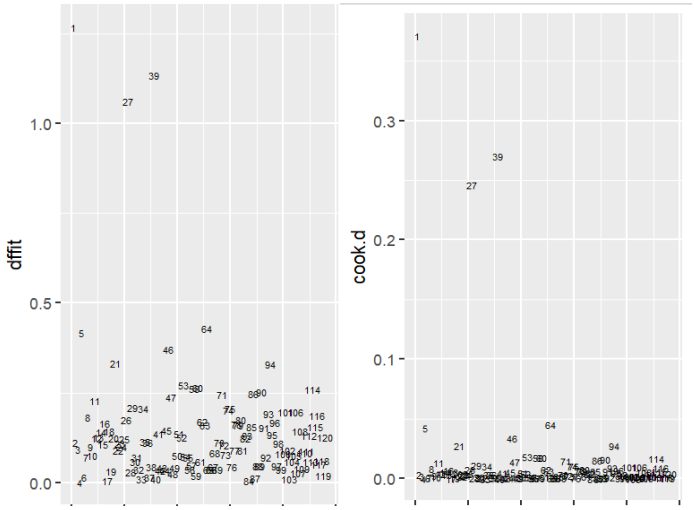


Figure Q4

(a) Explain what is influence in regression analysis and how do influential points affect your model.

(3 marks)

(b) Explain in your own words that is understandable to a sophisticated but non-technical audience without background in statistics (for example, the senior management in your company),

1. the purpose of influence analysis; and
2. the meaning of the 5 influence measure above; and
3. the next steps after the influence analysis.

(8 marks)

(c) Perform an analysis of influence, showing your working and justification clearly. Identify potential influential points.

(7 marks)

(d) One crude measure to manage outliers is to remove the data points. Discuss the pros and cons of this approach.

(4 marks)

(e) What are some strategies which could be implemented to handle the influential points instead of removing them?

(3 marks)

----- END OF PAPER -----